M3503 WEEK 10 2022

Solutions About Regular Singular Points

A singular point x_0 of $y'' + P(x)y' + Q(x)y = 0$ is a regular singular point if both $(x-x_0)P(x)$ and $(x-x_0)^2 Q(x)$ are analytic at x_0. Otherwise, x_0 is an irregular singular point.

Examples
Determine the singular points and classify them as regular or irregular

① $(x^2-9)^2 y'' + (x+3)y' + 2y = 0$

② $x^3(x^2-25)(x-2)^2 y'' + 3x(x-2)y' + 7(x+5)y = 0$

soln

① $x^2 - 9 = 0 \Rightarrow x = \pm 3$

For $x = -3$

$\frac{(x+3)(x+3)}{(x^2-9)^2} = \frac{(x+3)^2}{(x-3)^2(x+3)^2} = \frac{1}{(x-3)^2}$ which is analytic at $x = -3$

$\frac{2(x+3)^2}{(x^2-9)^2} = \frac{2}{(x-3)^2}$ which is analytic at $x = -3$

$\therefore x = -3$ is a regular singular point

For $x = 3$

$\frac{(x+3)(x-3)}{(x^2-9)^2} = \frac{1}{x^2-9}$ which is not analytic at $x = 3$

$\therefore x = 3$ is an irregular singular point.

Q. $x^3(x^2-25)(x-2)^2 y'' + 3x(x-2)y' + 7(x+5)y = 0$

$x^3(x^2-25)(x-2)^2 = 0$

$\Rightarrow x = 0, \pm 5, 2$

for $x = 0$

$\frac{3x^2(x-2)}{x^3(x^2-25)(x-2)^2} = \frac{3(x-2)}{x(x^2-25)(x-2)^2}$ which is not analytic at $x=0$

$\therefore x=0$ is an irregular singular point

for $x = 5$

$\frac{3x(x-2)(x-5)}{x^3(x^2-25)(x-2)^2} = \frac{3}{x(x+5)(x-2)}$ which is analytic at $x=5$

$\frac{7(x+5)(x-5)^2}{x^3(x^2-25)(x-2)^2} = \frac{7(x-5)}{x^3(x-2)^2}$ which is analytic at $x=5$

$\therefore x=5$ is a regular singular point

For $x = -5$

$\frac{3x(x-2)(x+5)}{x^3(x^2-25)(x-2)^2} = \frac{3}{x^2(x-5)(x-2)}$ which is analytic at $x=-5$

$\frac{7(x+5)^3}{x^3(x^2-25)(x-2)^2} = \frac{7(x+5)^2}{x^3(x-5)(x-2)^2}$ which is analytic at $x=-5$

$\therefore x=-5$ is a regular singular point

for $x = 2$

$\frac{3x(x-2)^2}{x^3(x^2-25)(x-2)^2} = \frac{3}{x^2(x^2-25)}$ which is analytic at $x=2$

$\frac{7(x+5)(x-2)^2}{x^3(x^2-25)(x-2)^2} = \frac{7}{x^3(x-5)}$ which is analytic at $x=2$

$\therefore x=2$ is a regular singular point.

FROBENIUS THEOREM

If x_0 is a regular singular point of the equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, then there exists at least one solution of the form $y = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r}$, where r, known as the indicial root or exponent of the singularity x_0, is a constant to be determined

INDICIAL EQUATION

If x_0 is a regular singular point of $y'' + P(x)y' + Q(x)y = 0$, then the indicial equation for this point is

$r(r-1) + P_0 r + Q_0 = 0$, where

$P_0 = \lim_{x \to x_0} (x-x_0) P(x)$ and

$Q_0 = \lim_{x \to x_0} (x-x_0)^2 Q(x)$

METHOD OF FROBENIUS

To find a series solution about a regular singular point x_0 of $y'' + P(x)y' + Q(x)y = 0$, substitute $y = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r}$ into the equation, determine the indicial roots and then determine the coefficients corresponding to each root.

Suppose x_0 is a regular singular point of $y'' + P(x)y' + Q(x)y = 0$ and that the indicial roots r_1 and r_2 are real.

Case I If $r_1 > r_2$ and $r_1 - r_2$ is not an integer, then there exist two linearly independent solutions of the form $y_1 = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r_1}$ and $y_2 = \sum_{n=0}^{\infty} b_n (x-x_0)^{n+r_2}$,

$c_0 \neq 0$ and $b_0 \neq 0$

Case II If $r_1 > r_2$ and $r_1 - r_2$ is an integer, then there exist two linearly independent solutions of the form $y_1 = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r_1}$ and , $c_0 \neq 0$

$$y_2 = C y_1 \ln(x) + \sum_{n=0}^{\infty} b_n (x-x_0)^{n+r_2} \quad , b_0 \neq 0.$$

where C is a constant that could be zero.

Case III If $r_1 = r_2$, then the exists two linearly independent solutions of the form

$$y_1 = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+r_1} \quad , c_0 \neq 0$$

$$y_2 = y_1 \ln(x) + \sum_{n=1}^{\infty} b_n (x-x_0)^{n+r_2} \quad , b_0 \neq 0$$

When $r_1 - r_2$ is an integer, we may or may not find two solutions. When $r_1 = r_2$, the method of Frobenius fails to give the second solution.

One more way to obtain the second solution is to use the fact that $y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx$

is also a solution to $y'' + P(x) y' + Q(x) y = 0$

Examples

Find the series solution about $x = 0$

① $2xy'' - y' + 2y = 0$

② $xy'' + 2y' - xy = 0$

③ $xy'' + y = 0$

④ $x^2y'' - xy' + (1-x)y$

Soln

① Let $y = \sum_{n=0}^{\infty} C_n x^{n+r}$, substituting into the equation we get

$$2x\sum_{n=0}^{\infty}(n+r)(n+r-1)C_n x^{n+r-2} - \sum_{n=0}^{\infty}(n+r)C_n x^{n+r-1} + 2\sum_{n=0}^{\infty}C_n x^{n+r} = 0$$

$$\Rightarrow 2r(r-1)C_0 x^{r-1} + 2\sum_{n=1}^{\infty}(n+r)(n+r-1)C_n x^{n+r-1} - rC_0 x^{r-1} - \sum_{n=1}^{\infty}(n+r)C_n x^{n+r-1} + 2\sum_{n=0}^{\infty}C_1 x^{n+r} = 0$$

$$\Rightarrow [2(r-1)-1]rC_0 x^{r-1} + 2\sum_{n=0}^{\infty}(n+1+r)(n+r)C_{n+1}x^{n+r} - \sum_{n=0}^{\infty}(n+1+r)C_{n+1}x^{n+r} + 2\sum_{n=0}^{\infty}C_n x^{n+r} = 0$$

Comparing coefficients, we get

$$[2(r-1)-1]r = 0$$

$$\Rightarrow r = 3/2 \quad \text{or} \quad r = 0$$

and

$$2(n+1+r)(n+r)C_{n+1} - (n+1+r)C_{n+1} - 2C_n = 0$$

$$\Rightarrow C_{n+1} = \frac{-2C_n}{(n+1+r)(2(n+r)-1)}, \quad n = 0, 1, 2, \cdots$$

For $r = 0$

$$C_{n+1} = \frac{-2C_n}{(n+r)(2n-1)}, \quad n = 0, 1, 2, \cdots$$

$n=0$: $C_1 = 2C_0$

$n=1$: $C_2 = \frac{-2C_1}{2} = -2C_0$

$n=2$: $C_3 = \frac{-2C_2}{3(3)} = \frac{2(2)C_0}{3(3)} = \frac{2^3 C_0}{3 \cdot 3!}$

$n=3$: $C_4 = \frac{-2C_3}{4(5)} = \frac{-2(2)(2)C_0}{5(4)(3)(3)} = \frac{-2^4 C_0}{5 \cdot 3 \cdot 4!}$

$n=4$: $C_5 = \frac{-2C_4}{5(7)} = \frac{2(2)(2)(2)C_0}{7(5)(5)(4)(3)(3)} = \frac{2^5 C_0}{7 \cdot 5 \cdot 3 \cdot 5!}$

$$y_1 = \sum_{n=0}^{\infty} C_n x^n$$
$$= C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + \cdots$$
$$= C_0 \left[1 + 2x - 2x^2 + \frac{2^3}{3 \cdot 3!} x^3 - \frac{2^4}{5 \cdot 3 \cdot 4!} x^4 + \frac{2^5}{7 \cdot 5 \cdot 3 \cdot 5!} x^5 + \cdots\right]$$

For $r = 3/2$

$$C_{n+1} = \frac{-2C_n}{2(n+5/2)(n+1)} = \frac{-2C_n}{(2n+5)(n+1)}, \quad n = 0, 1, 2, \ldots$$

$n=0$: $C_1 = \frac{-2C_0}{5}$

$n=1$: $C_2 = \frac{-2C_1}{7 \cdot 2} = \frac{2^2 C_0}{7 \cdot 5 \cdot 2}$

$n=2$: $C_3 = \frac{-2C_2}{9 \cdot 3} = \frac{-2^3 C_0}{9 \cdot 7 \cdot 5 \cdot 3!}$

$n=3$: $C_4 = \frac{-2C_3}{11 \cdot 4} = \frac{2^4 C_0}{11 \cdot 9 \cdot 7 \cdot 5 \cdot 4!}$

$$y_2 = \sum_{n=0}^{\infty} C_n x^{n+3/2}$$

$$= x^{3/2} \sum_{n=0}^{\infty} C_n x^n$$

$$= x^{3/2}\left[C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \cdots\right]$$

$$= x^{3/2} C_0 \left[1 - \frac{2}{5}x + \frac{2^2}{7\cdot 5\cdot 2}x^2 - \frac{2^3}{9\cdot 7\cdot 5\cdot 3!}x^3 + \frac{2^4}{11\cdot 9\cdot 7\cdot 5\cdot 4!}x^4 \cdots\right]$$

$$y = A\left[1 + 2x - 2x^2 + \frac{2^3}{3\cdot 3!}x^3 - \frac{2^4 x^4}{5\cdot 3\cdot 4!} + \frac{2^5}{7\cdot 5\cdot 3\cdot 5!}x^5 - \cdots\right]$$
$$+ Bx^{3/2}\left[1 - \frac{2}{5}x + \frac{2^2}{7\cdot 5\cdot 2}x^2 - \frac{2^3}{9\cdot 7\cdot 5\cdot 3!}x^3 + \frac{2^4}{11\cdot 9\cdot 7\cdot 5\cdot 4!}x^4 - \cdots\right]$$

② $xy'' + 2y' - xy = 0$

Let $y = \sum_{n=0}^{\infty} C_n x^{n+r}$, substitution gives

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)C_n x^{n+r-1} + 2\sum_{n=0}^{\infty}(n+r)C_n x^{n+r-1} - \sum_{n=0}^{\infty} C_n x^{n+r+1} = 0$$

$$\Rightarrow r(r-1)C_0 x^{r-1} + (1+r)rC_1 x^r + \sum_{n=2}^{\infty}(n+r)(n+r-1)C_n x^{n+r-1} + 2rC_0 x^{r-1}$$
$$+ 2(1+r)C_1 x^r + 2\sum_{n=2}^{\infty}(n+r)C_n x^{n+r-1} - \sum_{n=0}^{\infty} C_n x^{n+r+1} = 0$$

$$\Rightarrow r(r+1)C_0 x^{r-1} + (1+r)(2+r)C_1 x^r + \sum_{n=1}^{\infty}(n+1+r)(n+r)C_{n+1}x^{n+r}$$
$$+ 2\sum_{n=1}^{\infty}(n+r+1)C_{n+1}x^{n+r} - \sum_{n=1}^{\infty} C_{n-1}x^{n+r} = 0$$

Comparing the coefficients

x^{r-1}: $r(1+r)C_0 = 0$, since $C_0 \neq 0$, we have

$r = 0$ or $r = -1$

x^r: $(1+r)(2+r)C_1 = 0$

x^{n+r}: $(n+1+r)(n+r)C_{n+1} + 2(n+r+1)C_{n+1} - C_{n-1} = 0$

$$\Rightarrow C_{n+1} = \frac{C_{n-1}}{(n+1+r)(n+r+2)}, \quad n = 1, 2, 3, \cdots$$

For $r=0$,

$$c_1 = 0$$

and $$c_{n+1} = \frac{c_{n-1}}{(n+1)(n+2)}, \quad n = 1, 2, \ldots$$

$n=1$: $$c_2 = \frac{c_0}{2\cdot 3} = \frac{c_0}{3!}$$

$n=2$: $$c_3 = \frac{c_1}{3(4)} = 0$$

$n=3$: $$c_4 = \frac{c_2}{4\cdot 5} = \frac{c_0}{5!}$$

$n=4$: $$c_5 = \frac{c_3}{5\cdot 6} = 0$$

$n=5$: $$c_6 = \frac{c_4}{6\cdot 7} = \frac{c_0}{7!}$$

$$y_1 = \sum_{n=0}^{\infty} c_n x^n$$
$$= c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$$
$$= c_0 \left[1 + \frac{1}{3!}x^2 + \frac{1}{5!}x^4 + \frac{1}{7!}x^6 + \cdots\right]$$
$$= c_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n+1)!}$$

For $r = -1$

The equation $(1+r)(2+r)c_1 = 0$ does not tell us anything about c_1, so c_1 is still unknown.

$$c_{n+1} = \frac{c_{n-1}}{n(n+1)}, \quad n = 1, 2, 3, \ldots$$

$n=1$: $$c_2 = \frac{c_0}{2}$$

$n=2$: $$c_3 = \frac{c_1}{2(3)} = \frac{c_1}{3!}$$

$n=3: \; C_4 = \frac{C_2}{3\cdot 4} = \frac{C_0}{4!}$

$n=4: \; C_5 = \frac{C_3}{4(5)} = \frac{C_1}{5!}$

$n=5: \; C_6 = \frac{C_4}{5\cdot 6} = \frac{C_0}{6!}$

$$y_2 = x^{-1}\sum_{n=0}^{\infty} C_n x^n$$

$$= x^{-1}\left[C_0 + C_2x^2 + C_4x^4 + C_6x^6 + \cdots\right] + x^{-1}\left[C_1 + C_3x^3 + C_5x^5 + \cdots\right]$$

$$= x^{-1}C_0\left[1 + \tfrac{1}{2!}x^2 + \tfrac{1}{4!}x^4 + \tfrac{1}{6!}x^6 + \cdots\right] + x^{-1}C_1\left[x + \tfrac{1}{3!}x^3 + \tfrac{1}{5!}x^5 + \cdots\right]$$

$$= x^{-1}C_0\left[1 + \tfrac{1}{2!}x^2 + \tfrac{1}{4!}x^4 + \tfrac{1}{6!}x^6 + \cdots\right] + C_1\left[1 + \tfrac{1}{3!}x^2 + \tfrac{1}{5!}x^4 + \cdots\right]$$

$$= x^{-1}C_0\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!} + C_1\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n+1)!}$$

$$\therefore y = A\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n+1)!} + Bx^{-1}\sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!}$$

(3) $xy'' + y = 0$

Let $y = \sum_{n=0}^{\infty} C_n x^{n+r}$, substitution gives

$$\sum_{n=0}^{\infty}(n+r)(n+r-1)C_n x^{n+r-1} - \sum_{n=0}^{\infty} C_n x^{n+r} = 0$$

$$\Rightarrow r(r-1)C_0 x^{r-1} + \sum_{n=1}^{\infty}(n+r)(n+r-1)C_n x^{n+r-1} + \sum_{n=0}^{\infty} C_n x^{n+r} = 0$$

$$\Rightarrow r(r-1)C_0 x^{r-1} + \sum_{n=0}^{\infty}\left[(n+r+1)(n+r)C_{n+1} + C_n\right]x^{n+r} = 0 \quad \cdots (*)$$

$r(r-1) = 0 \Rightarrow r = 1$ or $r = 0$

$$C_{n+1} = -\frac{C_n}{(n+r+1)(n+r)} \quad , n = 0, 1, 2, \cdots$$

for $r = 0$

$$C_{n+1} = \frac{-C_n}{(n+1)n} \quad ; n = 1, 2, \cdots$$

for $r = 1$

$$C_{n+1} = \frac{-C_n}{(n+2)(n+1)} \quad , n = 0, 1, 2, \cdots$$

for $r=0$, since we cannot divide by zero, we go back to ⊛ and substitute $n = r = 0$, which gives $C_0 = 0$

$r=0$; $C_{n+1} = \frac{-C_n}{n(n+1)}$, $n = 1, 2, \cdots$

$$C_2 = -\frac{C_1}{2}$$

$$C_3 = -\frac{C_2}{3 \cdot 2} = \frac{C_1}{12}$$

$$C_4 = -\frac{C_3}{4 \cdot 3} = -\frac{C_1}{144}$$

$$y = \sum_{n=0}^{\infty} C_n x$$

$$= C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \cdots$$

$$= C_1 \left[x - \frac{1}{2} x^2 + \frac{1}{12} x^3 - \frac{1}{144} x^4 + \cdots \right]$$

$r=1$; $C_{n+1} = \frac{-C_n}{(n+2)(n+1)}$, $n = 0, 1, 2, \cdots$

$$C_1 = -\frac{C_0}{2}$$

$$C_2 = -\frac{C_1}{3 \cdot 2} = \frac{C_0}{12}$$

$$C_3 = -\frac{C_2}{4 \cdot 3} = -\frac{C_0}{144}$$

$$y = x \sum_{n=0}^{\infty} C_n x^n$$

$$= C_0 x + C_1 x^2 + C_2 x^3 + C_3 x^4 + \cdots$$

$$= C_0 \left[x - \frac{1}{2} x^2 + \frac{1}{12} x^3 - \frac{1}{144} x^4 + \cdots \right]$$

Thus we get just one solution.

$$y_1 = x - \frac{1}{2} x^2 + \frac{1}{12} x^3 - \frac{1}{144} x^4 + \cdots$$

The second solution is given by

$$y_2 = y_1 \int \frac{e^{-\int p \, dx}}{y_1^2} \, dx$$

$x y'' + y = 0 \Rightarrow y'' + \frac{y}{x} = 0 \equiv y'' + P(x)y' + Q(x) = 0$

$$y_2 = y_1 \int \frac{e^{-\int 0\,dx}}{(x - \frac{1}{2}x^2 + \frac{1}{12}x^3 - \frac{1}{144}x^4 + \cdots)^2}\,dx$$

$$= y_1 \int \frac{dx}{x^2 + x^3(-\frac{1}{2} - \frac{1}{2}) + x^4(\frac{1}{4} + \frac{1}{12} + \frac{1}{12}) + x^5(-\frac{1}{24} - \frac{1}{144} - \frac{1}{144} - \frac{1}{24}) + \cdots)}$$

$$= y_1 \int \frac{dx}{x^2 - x^3 + \frac{5}{12}x^4 - \frac{7}{72}x^5 + \cdots}$$

Using long division

$$\begin{array}{r|l} & 1 + x + \frac{7}{12} + \frac{19}{72}x + \cdots \\ \hline x^2 - x^3 + \frac{5}{12}x^4 - \frac{7}{72}x^5 + \cdots & 1 \\ & -(1 - x + \frac{5}{12}x^2 - \frac{7}{72}x^3 + \cdots) \\ \hline & x - \frac{5}{12}x^2 + \frac{7}{72}x^3 - \cdots \\ & -(x - x^2 + \frac{5}{12}x^3 - \frac{7}{72}x^4 + \cdots) \\ \hline & \frac{7}{12}x^2 - \frac{23}{72}x^3 + \frac{7}{72}x^4 - \cdots \\ & -(\frac{7}{12}x^2 - \frac{7}{12}x^3 + \frac{35}{144}x^4 - \frac{49}{864}x^5 + \cdots) \\ \hline & \frac{19}{72}x^3 + \cdots \\ & \frac{19}{72}x^3 + \cdots \end{array}$$

$$y_2 = y_1 \int \left(\frac{1}{x^2} + \frac{1}{x} + \frac{7}{12} + \frac{19}{72}x + \cdots\right) dx$$

$$= y_1 \left(\frac{-1}{x} + \ln x + \frac{7}{12}x + \frac{19}{144}x^2 + \cdots\right)$$

$$= y_1 \ln x + y_1 \left(\frac{-1}{x} + \frac{7}{12}x + \frac{19}{144}x^2 + \cdots\right)$$

$$y = A y_1 + B y_2$$

4. Exercise!!

Series Solution About A Point At Infinity

To obtain such a solution, we use the change of variables $z = \frac{1}{x}$ and expand about $z = 0$. The point at infinity is either an ordinary point, regular singular point or an irregular singular point if $z = 0$ is an ordinary, regular singular or irregular singular point respectively.

Examples

1. Show that infinity is a regular singular point of
$$x^3y'' - x^2y' - y = 0$$
2. Find the series solution of
$$x^3y'' - x^2y' - y = 0$$
about infinity

Sol^{n}

① Let $y(x) = V(\frac{1}{x})$

$$\Rightarrow y'(x) = -\frac{1}{x^2} V'(\tfrac{1}{x})$$

$$\Rightarrow y''(x) = \frac{2}{x^3} V'(\tfrac{1}{x}) + \frac{1}{x^4} V''(\tfrac{1}{x})$$

Putting $z = \frac{1}{x}$

$$y(x) = V(z)$$
$$y'(x) = -z^2 V'(z)$$
$$y''(x) = 2z^3 V'(z) + z^4 V''(z)$$

Substituting into the equation we get

$$\frac{1}{z^3}\left(2z^3 V' + z^4 V''\right) + \frac{1}{z^2}\left(z^2 V'\right) - V = 0$$
$$\Rightarrow 2V' + zV'' + V' - V = 0$$
$$\Rightarrow zV'' + 3V' - V = 0$$
$$\Rightarrow V'' + \frac{3V'}{z} - \frac{V}{z} = 0$$

Since $\frac{3}{z} \cdot z = 3$ and $-\frac{z^2}{z} = -z$ are both analytic at $z=0$, $z=0$ is a regular singular point, thus infinity is a regular singular point.

2. Hint: Find series solution of
$zV'' + 3V' - V$
about $z=0$ and substitute back $z = \frac{1}{x}$.